

Bluestock Mutual Fund Analysis Capstone Project

1. Executive Summary

This project analyzes mutual fund industry data from 2022–2025 using Python, SQLite, and Power BI. The objective was to build an end-to-end analytics solution covering data ingestion, cleaning, exploratory analysis, performance analytics, investor behavior analysis, dashboard development, and advanced risk analytics.

2. Project Objectives

- Build ETL pipeline
- Perform EDA
- Analyze fund performance
- Create Power BI dashboard
- Develop advanced analytics
- Generate actionable insights

3. Data Sources

Fund Master

Contains scheme information, categories, expense ratios, fund managers, and risk grades.

NAV History

Daily NAV records from 2022–2026.

Scheme Performance

Returns, alpha, beta, Sharpe ratio, Sortino ratio, drawdown, and AUM.

Investor Transactions

Investor demographics, SIPs, redemptions, and transaction behavior.

Portfolio Holdings

Sector allocations and portfolio weights.

Benchmark Indices

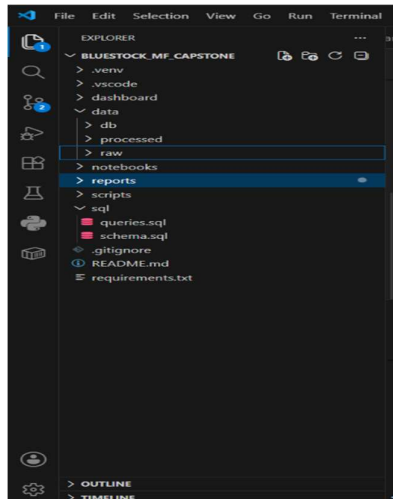
NIFTY50 and NIFTY100 historical values.

4. ETL Design

Explain:

- Data ingestion

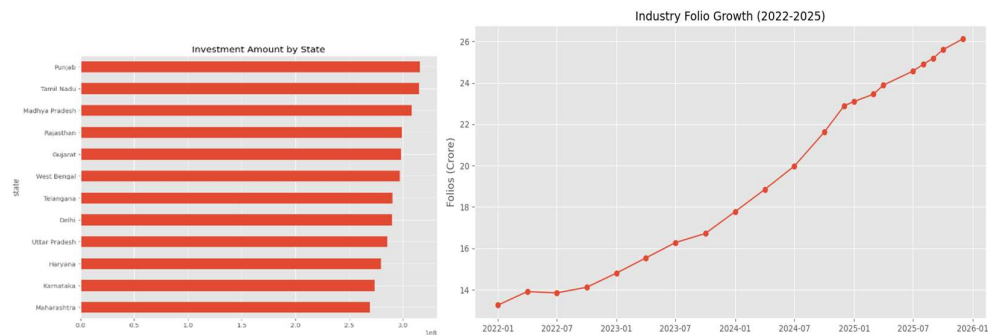
- Data cleaning
- Validation
- SQLite storage
- Analytics layer

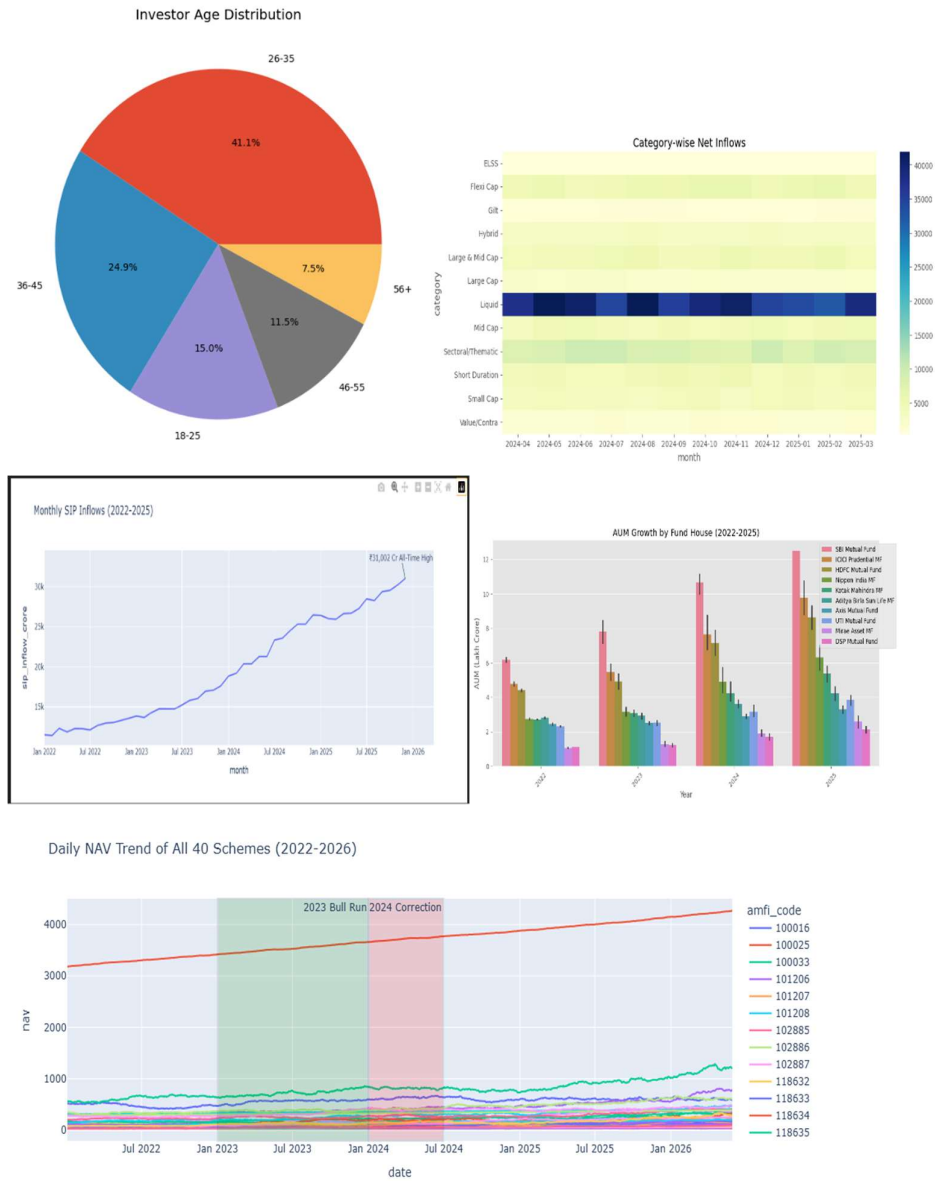


5. Exploratory Data Analysis

Include:

- NAV trend analysis
- AUM growth
- SIP inflow trends
- Category inflow heatmap
- Investor demographics
- Geographic distribution
- Folio growth



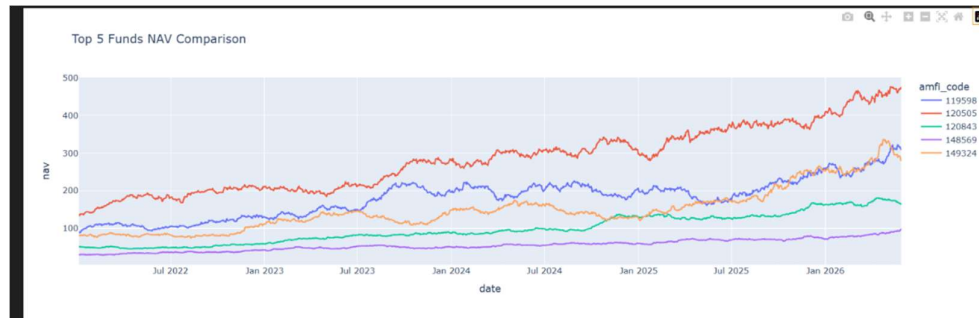


6. Performance Analytics

Include:

- CAGR analysis
- Sharpe ratio
- Sortino ratio
- Alpha and Beta

- Maximum drawdown
- Fund scorecard



7. Dashboard Overview

Insert screenshots:

- Industry Overview
- Fund Performance
- Investor Analytics
- SIP & Market Trends



8. Advanced Analytics

Include:

- VaR and CVaR
- Rolling Sharpe
- Investor Cohort Analysis
- SIP Continuity Analysis
- Sector HHI
- Fund Recommendation Engine

9. Key Findings

1. SIP inflows reached all-time highs.
2. Equity funds dominated investor interest.
3. High-Sharpe funds consistently outperformed.
4. Certain investors showed SIP discontinuity risk.
5. Portfolio concentration varied significantly across funds.

10. Limitations

- Sample dataset size
- Simplified recommendation engine
- Historical assumptions

11. Recommendations

- Expand dataset coverage
- Add ML-based recommendation models
- Deploy dashboard to cloud

12. Conclusion

The project successfully delivered a complete mutual fund analytics platform covering ETL, analytics, visualization, and reporting.